Pseudoinverse Guidance and Posterior Sampling for Diffusion-Based Image Restoration: A Theoretical and Empirical Comparison

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1 Introduction

Recovering a clean image from a degraded observation that is blurred, masked, downsampled, corrupted by noise, etc. is a fundamental problem in computational imaging. When the degradation process is known but ill-conditioned, infinitely many images may be consistent with the observation, and a prior over natural images is needed to select a plausible solution.

Diffusion models [1, 2] provide a powerful class of learned priors for this purpose. Two paradigms exist to apply them to inverse problems: training a *problem-specific* conditional model per degradation [3, 4], or using a single pretrained model in a *plug-and-play* fashion, guiding the reverse sampling toward measurement-consistent images at inference time [5, 6, 7]. The second approach is far more flexible but requires approximating the intractable conditional score $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t)$, which indicates how the current noisy iterate should be adjusted for consistency with the observation.

Diffusion Posterior Sampling (DPS) [6] and Pseudoinverse-Guided Diffusion Models (ΠGDM) [7] propose two different approximations to this score. DPS replaces the unknown clean image with the denoiser's current point estimate and differentiates the data-fidelity loss. ΠGDM instead models the denoiser's uncertainty as a Gaussian and analytically marginalises, yielding principled noise handling, an adaptive guidance weight, and the use of the pseudoinverse rather than the transpose of the measurement operator.

In this report, we trace each method from its probabilistic assumptions to its implementable algorithm, and evaluate both on inpainting and Gaussian deblurring of increasing difficulty, and on composed measurements (noiseless deblurring + random inpainting + 4× super-resolution) using a pretrained DDPM on FFHQ 256×256. Our experiments confirm the theoretical predictions: ΠGDM consistently outperforms DPS, with particularly large gains on deblurring and improved robustness to measurement noise.

2 Background

2.1 Diffusion Models and the DDPM Framework

Denoising Diffusion Probabilistic Models (DDPM) [1] define a forward noising process that gradually corrupts a clean image $\mathbf{x}_0 \sim p_0(\mathbf{x}_0)$ into Gaussian noise over T discrete timesteps. The forward transition is:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\ (1-\bar{\alpha}_t)\,\mathbf{I}\big) \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$ is the cumulative signal retention factor and $\{\beta_t\}_{t=1}^{T}$ is a predetermined variance schedule with $\beta_t \in]0,1[$.

The score function $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ is approximated by a neural network $\boldsymbol{\varepsilon}_\theta(\mathbf{x}_t, t)$ trained via the denoising score matching objective [8]. By Tweedie's formula [9], the minimum mean squared error (MMSE) estimate of $\mathbf{x}_0$ given $\mathbf{x}_t$ is:

$$\hat{\mathbf{x}}_t \ := \ \mathbb{E}[\mathbf{x}_0|\mathbf{x}_t] \ = \ \frac{1}{\sqrt{\bar{\alpha}_t}}\big(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\varepsilon}_\theta(\mathbf{x}_t, t)\big) \tag{2}$$

The DDPM reverse process samples $\mathbf{x}_{t-1}$ from $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{x}_t, t),\, \beta_t\, \mathbf{I})$, where

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\, \boldsymbol{\varepsilon}_\theta(\mathbf{x}_t, t)\right) \tag{3}$$

with $\alpha_t = 1 - \beta_t$.

2.2 From DDPM to DDIM

Denoising Diffusion Implicit Models (DDIM) [10] generalise the DDPM sampler by introducing a stochasticity parameter $\eta \in [0, 1]$. Given a sub-sampled schedule of timesteps $\{v_i\}_{i=0}^{N}$ with $v_0 = 0$ and $v_N = T$, the DDIM update from time $t = v_i$ to $s = v_{i-1}$ is:

$$\mathbf{x}_s = \sqrt{\bar{\alpha}_s}\, \hat{\mathbf{x}}_t + c_1\, \boldsymbol{\varepsilon} + c_2\, \boldsymbol{\varepsilon}_\theta(\mathbf{x}_t, t) \tag{4}$$

where $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and the coefficients are

$$c_1 = \eta \sqrt{\left(1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_s}\right) \frac{1-\bar{\alpha}_s}{1-\bar{\alpha}_t}}, \qquad c_2 = \sqrt{1 - \bar{\alpha}_s - c_1^2} \tag{5}$$

Setting $\eta = 0$ yields a deterministic ODE sampler, while $\eta = 1$ recovers DDPM-like stochasticity. The key advantage of DDIM is that it can operate with $N \ll T$ steps (e.g., 20–100 instead of 1000) while maintaining sample quality.

2.3 Conditional Sampling via Bayes' Rule

To sample from the posterior $p_t(\mathbf{x}_t|\mathbf{y})$, we use Bayes' rule to decompose the conditional score:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y}) = \underbrace{\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)}_{\text{unconditional score}} + \underbrace{\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t)}_{\text{likelihood score}} \tag{6}$$

The first term is approximated by the pretrained network $\boldsymbol{\varepsilon}_\theta$. The second term is intractable because

$$p_t(\mathbf{y}|\mathbf{x}_t) = \int p(\mathbf{y}|\mathbf{x}_0)\, p_t(\mathbf{x}_0|\mathbf{x}_t)\, d\mathbf{x}_0 \tag{7}$$

involves marginalising over $\mathbf{x}_0$ under the posterior $p_t(\mathbf{x}_0|\mathbf{x}_t)$, which itself requires running the full reverse diffusion process from t to 0.

This intractability is the point where DPS and ΠGDM diverge: they propose different approximations to $p_t(\mathbf{y}|\mathbf{x}_t)$ or equivalently to $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t)$.

3 Theoretical Comparison of DPS and ΠGDM

3.1 DPS: Diffusion Posterior Sampling

3.1.1 Score Approximation

DPS [6] replaces the intractable marginalisation in Eq. (7) with a point estimate. Specifically, it approximates $p_t(\mathbf{x}_0|\mathbf{x}_t) \approx \delta(\mathbf{x}_0 - \hat{\mathbf{x}}_t)$, collapsing the posterior to a Dirac mass at the Tweedie estimate. This yields:

$$p_t(\mathbf{y}|\mathbf{x}_t) \approx p(\mathbf{y}|\mathbf{x}_0)\big|_{\mathbf{x}_0 = \hat{\mathbf{x}}_t} \tag{8}$$

where $p(\mathbf{y}|\mathbf{x}_0) = \mathcal{N}(H\mathbf{x}_0, \sigma_y^2 \mathbf{I})$ follows from the linear Gaussian measurement model $\mathbf{y} = H\mathbf{x}_0 + \mathbf{z}$. Taking the gradient with respect to $\mathbf{x}_t$ gives:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t) \approx -\frac{1}{\sigma_y^2} \nabla_{\mathbf{x}_t} \|\mathbf{y} - H\hat{\mathbf{x}}_t\|_2^2 \tag{9}$$

For the general case $\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \mathbf{z}$, DPS uses $\nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_t)\|_2^2$ directly. The gradient $\nabla_{\mathbf{x}_t}$ involves backpropagation through the denoiser via the chain rule, since $\hat{\mathbf{x}}_t$ depends on $\mathbf{x}_t$ through Eq. (2).

3.1.2 Algorithm

DPS uses DDPM ancestral sampling (all $T = 1000$ steps) with the gradient correction added at each step:

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t}\,\mathbf{z} - \zeta_t \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_t)\|_2^2 \tag{10}$$

where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and the step size is $\zeta_t = \zeta'/\|\mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_t)\|$ with ζ' a task-dependent hyperparameter. This normalisation heuristic ensures the gradient correction has a roughly constant magnitude regardless of the residual norm.

3.2 ΠGDM: Pseudoinverse-Guided Diffusion Models

3.2.1 Score Approximation

ΠGDM [7] replaces the intractable posterior $p_t(\mathbf{x}_0|\mathbf{x}_t)$ not with a point mass but with a Gaussian:

$$p_t(\mathbf{x}_0|\mathbf{x}_t) \approx \mathcal{N}(\hat{\mathbf{x}}_t,\ r_t^2\,\mathbf{I}) \tag{11}$$

where $\hat{\mathbf{x}}_t$ is the Tweedie estimate (Eq. 2) and r_t is a time-dependent standard deviation. This approximation can be motivated as the optimal Gaussian variational approximation, since minimising $\mathbb{E}_{p(\mathbf{x}_0,\mathbf{x}_t)}[\mathrm{KL}(p(\mathbf{x}_0|\mathbf{x}_t)\|q(\mathbf{x}_0|\mathbf{x}_t))]$ over Gaussians q with fixed variance recovers $\hat{\mathbf{x}}_t$ as the optimal mean [7].

Since both $p_t(\mathbf{x}_0|\mathbf{x}_t)$ and $p(\mathbf{y}|\mathbf{x}_0) = \mathcal{N}(H\mathbf{x}_0, \sigma_y^2\mathbf{I})$ are Gaussian (or approximated as such), the marginalisation in Eq. (7) can be performed in closed form:

$$p_t(\mathbf{y}|\mathbf{x}_t) \approx \mathcal{N}\big(H\hat{\mathbf{x}}_t,\ r_t^2\,HH^\top + \sigma_y^2\,\mathbf{I}\big) \tag{12}$$

Taking the score gives the central formula of ΠGDM:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t) \approx \left((\mathbf{y} - H\hat{\mathbf{x}}_t)^\top \big(r_t^2\,HH^\top + \sigma_y^2\,\mathbf{I}\big)^{-1} H\,\frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t}\right)^\top \tag{13}$$

This is a vector-Jacobian product (VJP), computable by a single backward pass through the denoiser.

The variance parameter r_t is derived from the closed-form posterior when $p_0(\mathbf{x}_0) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, yielding [7]:

$$r_t = \sqrt{\frac{\sigma_t^2}{\sigma_t^2 + 1}}, \qquad \text{where } \sigma_t^2 = \frac{1 - \bar{\alpha}_t}{\bar{\alpha}_t} \tag{14}$$

This gives $r_0 = 0$ (no uncertainty at $t = 0$) and $r_T \approx 1$ (maximum uncertainty at $t = T$). The parameter r_t provides an implicit noise-aware regularisation: when $\sigma_t \ll \sigma_y$ (late timesteps, near-clean signal), $r_t^2/(r_t^2 + \sigma_y^2) \approx \sigma_t^2/\sigma_y^2 \ll 1$, so the guidance term is downweighted and the sampler avoids overfitting the measurement noise. When σ_t is large (early timesteps), $r_t \approx 1$ and the guidance follows the measurement freely.

3.2.2 Noiseless Case and the Pseudoinverse

When $\sigma_y = 0$, Eq. (13) simplifies. The covariance becomes $r_t^2\,HH^\top$, so $(r_t^2 HH^\top)^{-1} = r_t^{-2}(HH^\top)^{-1}$ and

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t) \approx r_t^{-2} \left((H^\dagger\mathbf{y} - H^\dagger H\hat{\mathbf{x}}_t)^\top \frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t}\right)^\top \tag{15}$$

where $H^\dagger = H^\top(HH^\top)^{-1}$ is the Moore–Penrose pseudoinverse. As we discuss in Section 3.2.4, the r_t^{-2} factor cancels with the guidance weight, so the final update involves no explicit r_t dependence.

Crucially, this formulation extends to non-linear and even non-differentiable measurement models. For a general operator h, one defines a generalised pseudoinverse $h^\dagger$ satisfying $h(h^\dagger(h(\mathbf{x}))) = h(\mathbf{x})$, and the guidance becomes [7]:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t) \approx r_t^{-2} \left((h^\dagger(\mathbf{y}) - h^\dagger(h(\hat{\mathbf{x}}_t)))^\top \frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t}\right)^\top \tag{16}$$

This allows ΠGDM to handle operators such as JPEG compression or quantisation, where h is not differentiable. The differentiation only passes through the denoiser $\hat{\mathbf{x}}_t$, not through h or $h^\dagger$.

3.2.3 Noisy Case

For $\sigma_y > 0$, the guidance vector $\boldsymbol{v}$ entering the VJP is:

$$\boldsymbol{v} = H^\top \left(HH^\top + \frac{\sigma_y^2}{r_t^2} \mathbf{I} \right)^{-1} (\mathbf{y} - H\hat{\mathbf{x}}_t) \tag{17}$$

The regularisation parameter σ_y^2/r_t^2 balances measurement fidelity against noise amplification. As $\sigma_y \to 0$, this converges to $H^\dagger(\mathbf{y} - H\hat{\mathbf{x}}_t)$, recovering the noiseless case. The expression can be computed efficiently for structured operators: for a diagonal H (inpainting), it reduces to element-wise division; for a circulant H (convolution), it can be evaluated in the Fourier domain as a frequency-wise filter.

3.2.4 Adaptive Weight Schedule

Unlike DPS, which requires manual tuning of the step size ζ', ΠGDM uses an *adaptive* weight schedule that emerges naturally from the algebra. The one-step update (Eq. 10 of [7]) is:

$$\mathbf{x}_s = f(\mathbf{x}_t;\, s, t, \eta) + r_t^2 \, \nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t) \tag{18}$$

where f denotes the DDIM step. In the noiseless case, the score contains r_t^{-2} (Eq. 15), so $r_t^2 \cdot r_t^{-2} = 1$ and the guidance has no free parameter. In the noisy case, the r_t^2 prefactor is absorbed into the regularisation: factoring r_t^2 into the matrix inverse in Eq. (13) converts $r_t^2(r_t^2 HH^\top + \sigma_y^2\mathbf{I})^{-1}$ into $(HH^\top + \sigma_y^2/r_t^2\,\mathbf{I})^{-1}$, leaving no explicit scale factor.

The effective guidance weight is therefore implicitly time-dependent, without any hyperparameter to tune. This is in contrast to reconstruction guidance [11], which uses a fixed weight w_r that can be optimal for one task but cause instability on another.

3.2.5 Algorithm

The full ΠGDM update for VP-SDE (Algorithm 1 of [7]) is:

$$\mathbf{x}_s = \sqrt{\bar{\alpha}_s}\, \hat{\mathbf{x}}_t + c_1\, \boldsymbol{\varepsilon} + c_2\, \boldsymbol{\varepsilon}_\theta + \sqrt{\bar{\alpha}_t}\, g \tag{19}$$

where the first three terms constitute the standard DDIM update (Eq. 4) and g is the guidance term. The factor $\sqrt{\bar{\alpha}_t}$ in front of g is a correction for the VP-SDE parametrisation. The guidance is computed as a VJP:

$$g = \left(\boldsymbol{v}^\top \frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t} \right)^\top \tag{20}$$

In practice, the VJP is computed by defining $s = \boldsymbol{v}^\top \hat{\mathbf{x}}_t$ (with $\boldsymbol{v}$ detached from the computation graph) and calling `torch.autograd.grad(s, x)`. Detaching $\boldsymbol{v}$ ensures that gradients only flow through $\hat{\mathbf{x}}_t$ and its dependence on $\mathbf{x}_t$ via the denoiser.

3.2.6 Composed Measurements

For measurement models that decompose as $h = h_k \circ \cdots \circ h_1$, ΠGDM approximates the pseudoinverse as $h^\dagger \approx h_1^\dagger \circ \cdots \circ h_k^\dagger$. This allows solving complex inverse problems (e.g., blur + downsampling + masking) using a single pretrained model, by composing the elementary pseudoinverses.

3.3 Comparison

We now synthesise the key differences between DPS and ΠGDM.

Score approximation. DPS models $p_t(\mathbf{x}_0|\mathbf{x}_t)$ as a Dirac $\delta(\mathbf{x}_0 - \hat{\mathbf{x}}_t)$, reducing the marginalisation to a point evaluation of the likelihood. ΠGDM uses a Gaussian $\mathcal{N}(\hat{\mathbf{x}}_t, r_t^2\mathbf{I})$, enabling closed-form marginalisation. The Gaussian approximation is more principled: it acknowledges uncertainty in the denoiser's prediction and leads to a consistent approximation of $p_t(\mathbf{x}_0|\mathbf{x}_t)$ that does not depend on the measurement operator H. In contrast, reconstruction guidance (which DPS builds upon) makes $p_t(\mathbf{x}_0|\mathbf{x}_t)$ implicitly dependent on H, leading to inconsistencies when different operators are used.

Noise handling. DPS has no explicit mechanism to avoid overfitting measurement noise: the gradient $\nabla_{\mathbf{x}_t}\|\mathbf{y} - H\hat{\mathbf{x}}_t\|^2$ treats noisy measurements identically to clean ones. ΠGDM's σ_y^2/r_t^2 term inside the matrix inverse naturally downweights noisy measurements at late timesteps (small r_t), providing built-in regularisation.

Guidance weights. DPS requires manual tuning of ζ' per task (the optimal value differs between super-resolution and deblurring [6]). ΠGDM's adaptive schedule is parameter-free: the effective weight emerges from the cancellation of r_t^2 with the score's r_t^{-2} factor.

Base sampler and efficiency. DPS uses DDPM ancestral sampling and requires all $T = 1000$ steps for competitive results [7]. ΠGDM uses DDIM with timestep sub-sampling and produces good results with 20–100 steps, making it 10–50× faster in terms of number of function evaluations (though each evaluation is slightly more expensive due to the VJP).

Differentiability and linearity requirements. DPS requires differentiability of $\mathcal{A}$ for the gradient $\nabla_{\mathbf{x}_t}\|\mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_t)\|^2$. ΠGDM's noiseless case (Eq. 16) only differentiates through the denoiser, not through h or $h^\dagger$, allowing non-differentiable operators. However, ΠGDM's noisy case (Eq. 13) requires linearity of H.

4 Experiments

4.1 Experimental Setup

Dataset and model. We use the FFHQ dataset at resolution 256×256, with a validation set of 1000 images. The diffusion prior is a pretrained unconditional DDPM U-Net from [6], which uses the architecture of [12] with learned variance [13]. The same model is used for both DPS and ΠGDM.

Tasks. We evaluate on five tasks of increasing difficulty:

1. **Random inpainting (50% missing):** each pixel is independently masked with probability 0.5.
2. **Random inpainting (70% missing):** same as above with probability 0.7.
3. **Gaussian deblurring (clean):** circular convolution with a 9×9 Gaussian kernel ($\sigma_{\text{blur}} = 3$), no measurement noise.
4. **Gaussian deblurring (noisy):** same blur kernel with measurement noise $\sigma_y \in \{0.02, 0.05, 0.10\}$.
5. **Composed measurements:** clean deblurring + random inpainting (50% and 70% missing) + 4× super-resolution.

Metrics. We report PSNR (higher is better) and SSIM (higher is better). For noisy deblurring, metrics are averaged over 30 images.

Hyperparameters. Both methods are tuned via grid search for fair comparison. For DPS: $\zeta' \in \{0.03, 0.05, 0.10, 0.20\}$. For ΠGDM: number of steps $N \in \{50, 100\}$ and $\eta \in \{0, 0.5, 1.0\}$; for noisy deblurring, the measurement noise level σ_y is additionally provided to the guidance term. Each method is reported with its best configuration under the same validation protocol.

4.2 Results

4.2.1 Inpainting

Table 1 summarises the quantitative results on random inpainting with 50% and 70% missing pixels. ΠGDM outperforms DPS by a substantial margin in both settings: +6.08 dB PSNR and +0.105 SSIM at 50% missing, and +4.88 dB PSNR and +0.118 SSIM at 70% missing.

For inpainting, $H = \text{diag}(\boldsymbol{m})$ is a binary diagonal matrix, so $H^\top = H$ and $H^\dagger = H$. Therefore, the pseudoinverse and transpose coincide, and the theoretical advantage of ΠGDM over DPS does not come from the pseudoinverse itself but from the DDIM sampler (enabling fewer, more effective steps), the adaptive weight schedule, and the overall structure of the guidance term.

Table 1: Random inpainting results (single representative image).

	50% missing		70% missing	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑
DPS	31.42	0.848	29.75	0.803
ΠGDM	**37.50**	**0.953**	**34.63**	**0.921**

Both methods degrade as the missing ratio increases, but ΠGDM maintains a larger advantage in the harder 70% setting, suggesting better robustness to increasingly ill-posed problems.

4.2.2 Gaussian Deblurring (Clean)

Table 2 shows results on noiseless Gaussian deblurring. The performance gap is dramatic: ΠGDM achieves +13.60 dB PSNR and +0.252 SSIM over DPS.

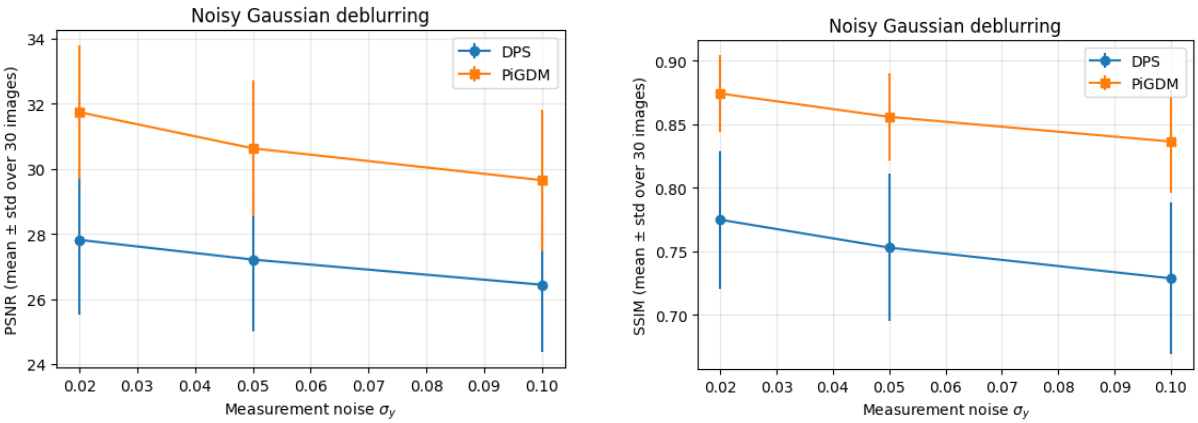
Table 2: Gaussian deblurring with clean measurements (single representative image).

Method	PSNR ↑	SSIM ↑
DPS	28.06	0.725
ΠGDM	**41.66**	**0.977**

This large gap is consistent with the theory. The Gaussian blur operator has highly non-uniform singular values [7]: low-frequency components are preserved while high-frequency ones are severely attenuated. The pseudoinverse $H^\dagger$ in ΠGDM treats all non-zero singular directions equally (via $H^\dagger H = V^\top \mathbb{I}(\Sigma^2)V$), whereas the transpose-based correction in DPS rescales them by Σ^2 (via $H^\top H = V^\top \Sigma^2 V$), amplifying the imbalance and producing less accurate guidance for the attenuated frequencies.

4.2.3 Gaussian Deblurring (Noisy)

Figure 1 shows PSNR and SSIM as a function of measurement noise σ_y, averaged over 30 images. Both methods degrade as noise increases, but ΠGDM maintains a consistent advantage of approximately 3–4 dB in PSNR and 0.05–0.10 in SSIM across all noise levels.



(a) PSNR vs. σ_y (30-image average). **(b)** SSIM vs. σ_y (30-image average).

Figure 1: Noisy Gaussian deblurring: metrics vs. measurement noise level, averaged over 30 FFHQ validation images. ΠGDM consistently outperforms DPS and degrades more gracefully.

This confirms the theoretical prediction: ΠGDM's regularisation term (σ_y^2/r_t^2 in Eq. 17) naturally prevents overfitting noisy measurements, while DPS has no built-in mechanism for this. The smoother degradation curve of ΠGDM further illustrates its robustness.

4.2.4 Composed Measurements

We finally consider a composed linear measurement operator combining three degradations in a single task. Starting from a clean image $\mathbf{x}$, we first apply a clean Gaussian blur B, then a random inpainting mask M_ρ with missing ratio $\rho \in \{0.5, 0.7\}$, and finally a $4\times$ downsampling operator S_4. The forward model is therefore

$$\mathbf{y} = S_4 M_\rho B \mathbf{x} \tag{21}$$

In our implementation, the corresponding pseudoinverse is applied in reverse order,

$$H^\dagger \approx B_\varepsilon^\dagger \, M_\rho \, S_4^\dagger \tag{22}$$

where $S_4^\dagger$ is a nearest-neighbor upsampling, $M_\rho^\dagger = M_\rho$ since the mask is diagonal and idempotent, and $B_\varepsilon^\dagger$ is a regularized inverse blur. This regularization is important in practice because the inverse of the blur is ill-conditioned; without it, both the display of the pseudoinverted measurement and the PiGDM guidance may become numerically unstable.

Table 3 reports the results for the two masking ratios. PiGDM remains better than DPS in both settings, with gains of +6.19 dB and +0.213 SSIM at 50% masking, and +5.12 dB and +0.196 SSIM at 70% masking. As expected, performance decreases when the masking ratio increases, but the relative advantage of PiGDM is preserved. This is consistent with the main motivation of PiGDM: the method naturally extends to composed measurement models by composing the forward operators and their approximate pseudoinverses in reverse order.

Table 3: Composed measurement results: clean deblurring + random inpainting + 4× super-resolution (single representative image).

	50% missing		70% missing	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑
DPS	23.26	0.615	22.03	0.579
ΠGDM	**29.45**	**0.828**	**27.15**	**0.775**

The qualitative illustrations are reported at the end of this report. They show that DPS tends to produce smoother reconstructions, while PiGDM better preserves face structure, contrast, and local details. The difference becomes more visible in the harder 70% masking setting, where the composed degradation is more severe.

4.2.5 Summary

Table 4 provides a task-level overview. ΠGDM wins on every task, with the largest advantage on deblurring where the operator conditioning strongly favours pseudoinverse guidance.

Table 4: Task-level summary of the comparison.

Task	Better method	Main observation
Inpainting 50%	ΠGDM	Sharper, more faithful reconstructions
Inpainting 70%	ΠGDM	Larger advantage in harder setting
Deblurring (clean)	ΠGDM	Very large quantitative gain (+13.6 dB)
Deblurring (noisy)	ΠGDM	Better robustness to increasing noise
Composed (50%)	ΠGDM	+6.19 dB gain, preserves structure under combined degradations
Composed (70%)	ΠGDM	+5.12 dB gain, advantage maintained in harder setting

5 Conclusion

In this work, we have provided a theoretical and empirical comparison of DPS and ΠGDM, two plug-and-play methods for solving inverse problems with pretrained diffusion models. The theoretical analysis reveals that the methods differ primarily in how they approximate the intractable posterior $p_t(\mathbf{x}_0|\mathbf{x}_t)$: DPS uses a Dirac point estimate, while ΠGDM uses a Gaussian with calibrated variance r_t^2. This

seemingly modest difference has cascading consequences: ΠGDM obtains a closed-form marginalised likelihood that naturally incorporates measurement noise through regularisation, uses the pseudoinverse rather than the transpose for better-conditioned guidance on operators with non-uniform singular values, and yields an adaptive weight schedule with no free hyperparameters.

Our experiments on FFHQ at 256×256 confirm these theoretical predictions. ΠGDM outperforms DPS on all tested tasks, with the largest gains on Gaussian deblurring (+13.6 dB clean, 3–4 dB noisy) where the operator conditioning is most challenging. On inpainting, ΠGDM still holds a substantial advantage (+4.9 to +6.1 dB) attributable to the DDIM sampler and adaptive weighting. On composed measurements combining deblurring, random inpainting, and $4\times$ super-resolution, ΠGDM maintains gains of +5.1 to +6.2 dB, showing that the pseudoinverse composition strategy is well-suited for multi-operator degradations.

Several directions remain for future work. First, extending the noisy ΠGDM formulation to non-linear operators is an open problem, as the Gaussian marginalisation requires linearity. Second, the computational cost of both methods remains high, combining these guidance strategies with distilled or consistency models could significantly accelerate inference. Finally, a systematic study on other degradations (motion blur, JPEG, etc.) would further clarify the practical scope of each method.

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A Visual Comparisons



Figure 2: Representative images from the FFHQ 256×256 validation set used in our experiments.

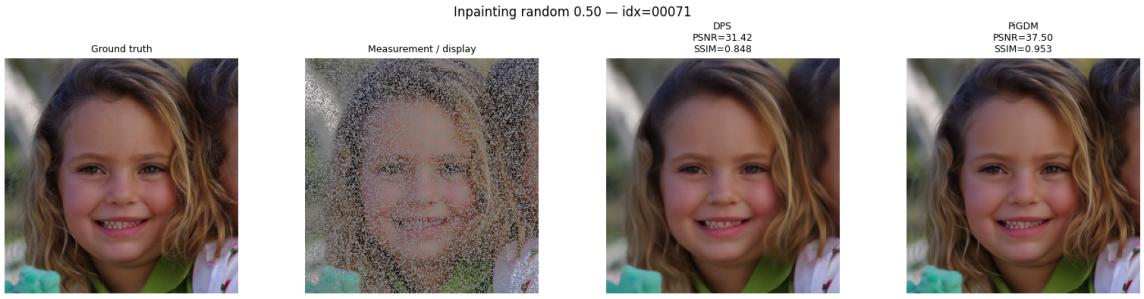


Figure 3: Random inpainting with 50% missing pixels. Left to right: ground truth, masked observation, DPS reconstruction, ΠGDM reconstruction.

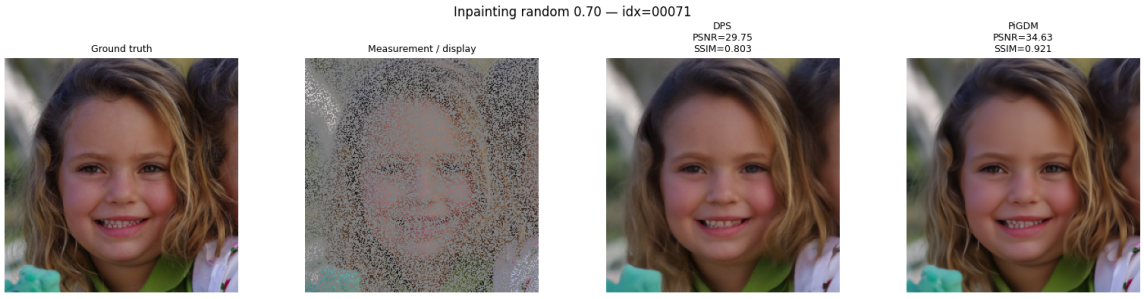


Figure 4: Random inpainting with 70% missing pixels. Left to right: ground truth, masked observation, DPS reconstruction, ΠGDM reconstruction.

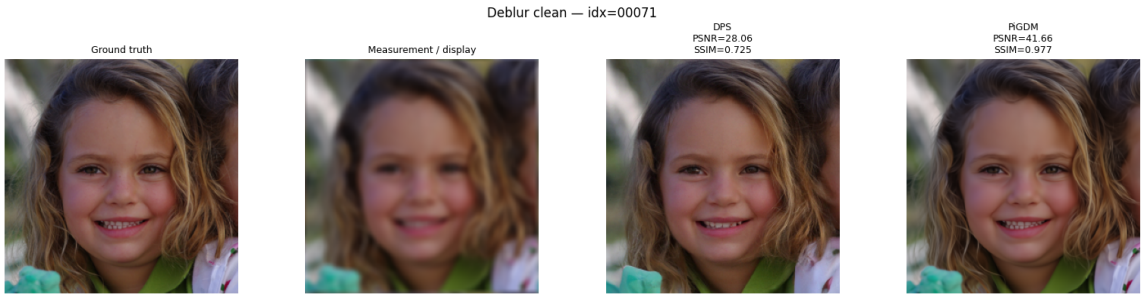


Figure 5: Gaussian deblurring with clean measurements. Left to right: ground truth, blurred observation, DPS reconstruction, ΠGDM reconstruction.

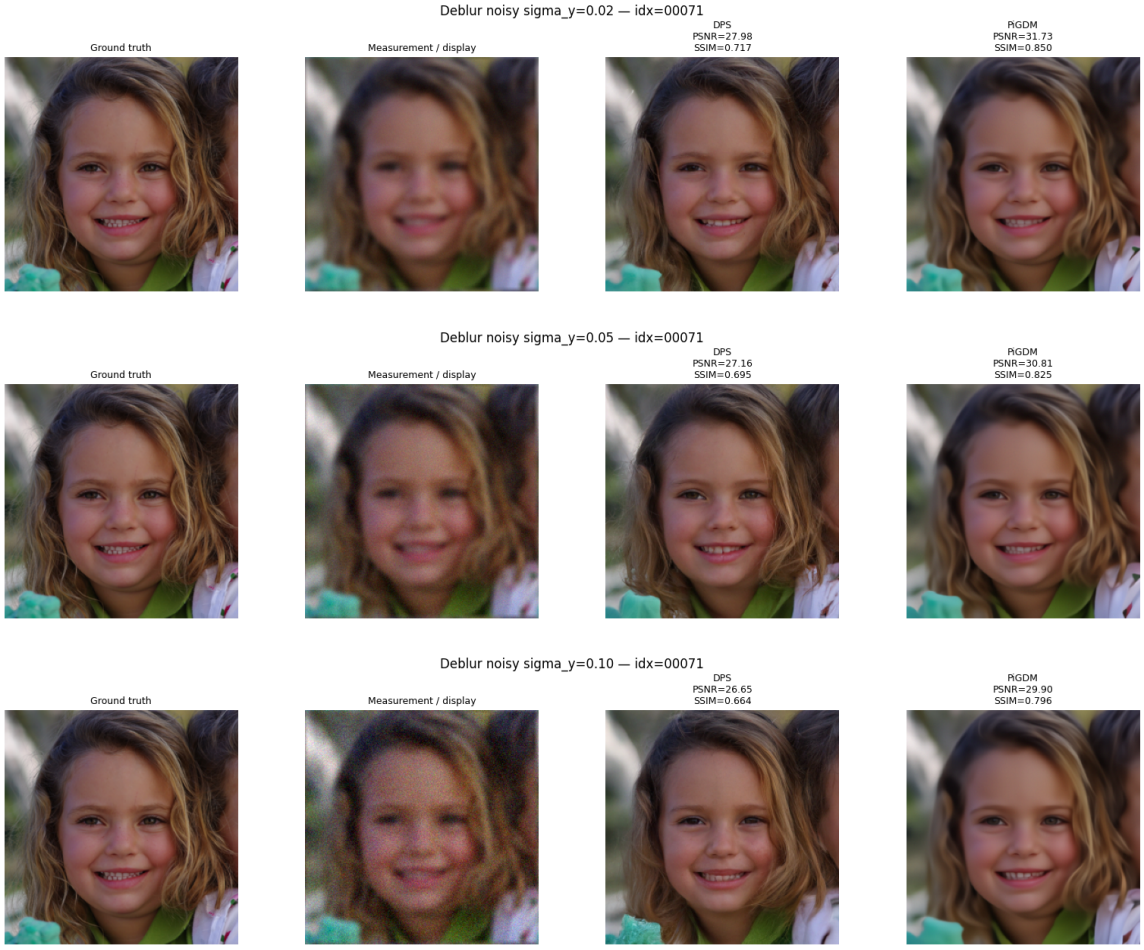


Figure 6: Noisy Gaussian deblurring at increasing noise levels: $\sigma_y = 0.02$ (top), $\sigma_y = 0.05$ (middle), $\sigma_y = 0.10$ (bottom). Left to right: ground truth, noisy blurred observation, DPS reconstruction, ΠGDM reconstruction.



Figure 7: Qualitative comparison for the composed measurement task *clean deblurring + random inpainting + 4× super-resolution*. Top: masking ratio 50%. Bottom: masking ratio 70%. In both cases, ΠGDM produces sharper and more faithful reconstructions than DPS.